

Manoj AI-Powered Learning Management System (AI-LMS)

1.1 Project Overview

The AI-Powered Learning Management System (AI-LMS) is an intelligent educational platform that combines Artificial Intelligence with modern learning technologies to deliver a personalized, interactive and scalable learning experience. The platform consists of four integrated applications: the Student Learning Portal (lmsfrontendai), the Administration Portal (lmsadmin), the Backend Services (lmsbackendai) and the AI Processing Engine (ai). Together these modules automate course creation, content management, AI-assisted learning, learner progress tracking and personalized recommendations.

1.2 Project Objectives

- Develop an AI-powered Learning Management System.
- Provide personalized learning through an AI Mentor.
- Improve concept understanding using a Knowledge Graph.
- Deliver AI-generated summaries and quizzes.
- Track learner progress and resume learning.
- Simplify course management through a centralized Admin Portal.
- Build a scalable architecture for future AI enhancements.

1.3 Proposed System

Administrators manage courses, videos, transcripts, quizzes and users from the Admin Portal. Backend APIs validate and persist all data. The learner portal dynamically loads published content and the AI Processing Engine provides transcript analysis, contextual tutoring, lesson summaries and recommendations.

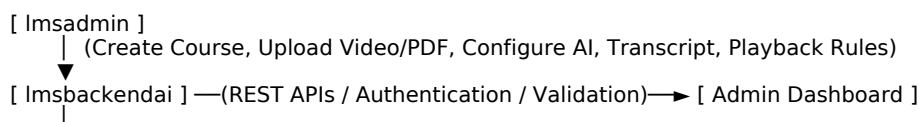
1.4 Key Features

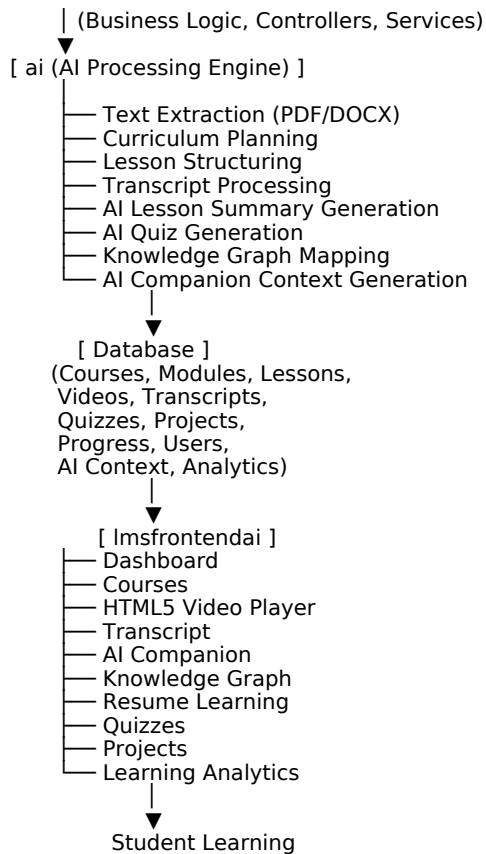
AI Mentor: Personalized tutor aware of lesson context and learner progress. Knowledge Graph: Visual roadmap connecting concepts and prerequisites. AI Companion: Context-aware assistant synchronized with the current lesson. Resume Learning: Continue exactly where the learner stopped. AI Lesson Summaries: Automatic revision notes. AI Quiz Generation: Dynamic assessments from lesson content. Learning Analytics: Engagement, completion and performance insights.

1.5 Expected Outcomes

Increase learner engagement, improve conceptual understanding, automate course creation tasks, provide personalized AI guidance, support continuous learning with progress tracking and establish a scalable AI-native LMS architecture.

2. End-to-End Architecture Flow





3. Integration Workflow

Administrator creates or updates content in Imsadmin. The request is processed by Imsbackendai through authenticated REST APIs. The backend validates the request, stores metadata in the database and triggers AI processing when required. Published content is automatically retrieved by Imsfrontendai without hardcoded data. During playback, the AI Companion receives the current lesson, timestamp, transcript and learner progress to provide contextual assistance. Progress, notes, bookmarks, quiz results and analytics are continuously synchronized back to the backend.